Use-Case Flow

Chat Application WebSocket Programming — Team 5, Section K

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Use Cases

Goal	**Create a new, unique user account.**
Actor	New User
Main flow	1. User opens the Chat Application. 2. System displays the Welcome screen with "Register" and "Login" options. 3. User selects the "Register" option. 4. System displays the registration form. 5. User provides all required registration details. 6. System validates the provided information. 7. System creates the new user account. 8. System displays a "Registration Successful" message.
Alternate Flow	Invalid or Duplicate Information At Step 6, if the registration details are invalid: 6a. System rejects the submission and displays a specific error message (e.g., "Username already taken," "Passwords do not match"). 6b. User corrects the details and returns to Main Flow Step 5.

Goal	**Securely access an existing account.**
Actor	Registered User
Main flow	1. User opens the application and selects "Login". 2. System displays the login form. 3. User provides their login credentials. 4. System validates the credentials. 5. Upon successful authentication, the system creates a user session. 6. System displays the user's main dashboard.
Alternate Flow	**Invalid Login Credentials** At Step 4, if login credentials are incorrect: 4a. System rejects the login and displays an error message. 4b. User retries the login and returns to Main Flow Step 3.

Goal	**Update personal profile information.**
Actor	User
Main flow	1. User logs into the system. 2. User navigates to the "Profile" section. 3. System displays the user's current profile information. 4. User selects an option to edit their profile. 5. User enters the updated details. 6. System validates the new information.

	7. System securely stores the updated profile. 8. System displays a confirmation message.
Alternate Flow	**Invalid Profile Information** At Step 6, if the provided information is invalid: 6a. System displays an error message explaining the issue. 6b. User corrects the details and returns to Main Flow Step 5.

Goal	**Recover access to an account with a forgotten password.**
Actor	User
Main flow	1. On the Login screen, User clicks the "Forgot Password?" link. 2. System displays a form asking for the user's registered email. 3. User enters their email and submits. 4. System validates that the email exists. 5. System sends a secure, time-limited password reset link to the email. 6. System displays a confirmation message.
Alternate Flow	**Email Not Found** At Step 4, if the email does not exist in the database: 4a. The system still displays the same confirmation message (Step 6) to prevent revealing which emails are registered.

Goal	**Add a new contact to the friend list.**
Actor	User

Main flow	1. User navigates to the "Add Friend" section. 2. User searches for another user by their username. 3. System displays the matching user profile. 4. User clicks the "Send Friend Request" button. 5. System sends a friend request notification to the recipient user. 6. System shows a "Friend request sent" confirmation.
Alternate Flow	**User Not Found** At Step 3, if no user matches the search criteria: 3a. System displays a "User not found" message. 3b. User refines their search and returns to Main Flow Step 2.

Goal	**Accept or decline a friend request.**
Actor	User
Main flow	1. User receives a notification for a new friend request. 2. User navigates to the "Friend Requests" page. 3. System displays a list of pending requests. 4. User chooses to "Accept" a request. 5. System adds each user to the other's contact list. 6. System removes the pending request.
Alternate Flow	**Decline Request** At Step 4, if the user chooses to "Decline" the request: 4a. System deletes the pending request. No relationship is created.

Goal	**Send a text message to a contact.**
Actor	User
Main flow	1. User selects a contact from their contact list. 2. System opens the one-to-one chat window. 3. User types a message into the input field. 4. User clicks the "Send" button. 5. System validates the message. 6. System sends the message to the server. 7. System updates the sender's chat window with the new message.
Alternate Flow	**Recipient is Offline** At Step 7, if the recipient is offline: 7a. The server stores the message. 7b. The message is delivered the next time the recipient logs in.

Goal	**Create a new group chat.**
Actor	User
Main flow	1. User selects the "Create Group" option. 2. System displays the user's contact list. 3. User selects two or more contacts. 4. User provides a name for the group. 5. System validates the group details. 6. System creates the new group chat with the creator as admin. 7. System opens the new group chat window.

Alternate Flow	**Insufficient Members** At Step 5, if the user selects fewer than the minimum required members: 5a. System displays an error: "A minimum of 2 contacts must be selected." 5b. User adds more members and returns to Main Flow Step 3.

Goal	**Manage members in a group chat.**
Actor	Group Admin
Main flow	1. Admin opens a group chat and navigates to "Group Settings". 2. System displays the member list and management options. 3. Admin selects "Add Members". 4. Admin chooses contacts to add. 5. System adds the new members to the group.
Alternate Flow	**Remove a Member** At Step 2, Admin can select a member and choose "Remove": 2a. System confirms the action. 2b. System removes the member from the group.

Goal	**Search for specific messages or contacts.**
Actor	User
Main flow	1. User clicks on the global search bar. 2. User enters a search query.

	3. System processes the query. 4. System displays a list of matching contacts and messages. 5. User selects a result to navigate to it.
Alternate Flow	**No Matching Results** At Step 4, if no results are found: 4a. System displays a "No results found" message.

Goal	**Block another user.**
Actor	User
Main flow	1. User navigates to another user's profile. 2. User selects the "Block User" option. 3. System displays a confirmation dialog. 4. User confirms the action. 5. System updates the relationship to "blocked," preventing future communication.
Alternate Flow	**Unblock a User** 1. User navigates to their "Blocked Users" list in Settings. 2. User selects a user and chooses "Unblock". 3. System restores communication ability.

Goal	**View message delivery and read status.**
Actor	User

Main flow	1. User sends a message. 2. System displays a "Sent" status indicator (e.g., one checkmark). 3. When the message is delivered, the indicator changes to "Delivered" (e.g., two checkmarks). 4. When the recipient views the message, the indicator changes to "Read" (e.g., two blue checkmarks).
Alternate Flow	**Read Receipts Disabled** At Step 4, if the recipient has disabled read receipts: 4a. The message status will not change from "Delivered."